

Day: _____

Date: _____

1, 5 $\rightarrow$ 10, 13 $\rightarrow$ 30,
(odd no) $\rightarrow$ Practice

Practice :-

i) $f(x) = e^{2x} ; [0, \ln 2]$

Sol. Since $e^{2x} \neq 0$, then
$$= \int_0^{\ln 2} e^{2x} dx$$

$$= \left. \frac{e^{2x}}{2} \right|_0^{\ln 2}$$

$$= \frac{e^{2 \ln 2}}{2} - \frac{e^0}{2}$$

$$= 2 - \frac{1}{2} \Rightarrow \boxed{3/2}$$

ii) $\int_{\sqrt{2}}^2 \frac{dx}{x \sqrt{x^2 - 1}}$

Sol

$$= \sec^{-1} |x| \Big|_{\sqrt{2}}^2$$

$$= \sec^{-1}(2) - \sec^{-1}(\sqrt{2})$$

$$= \frac{\pi}{3} - \frac{\pi}{4} \Rightarrow \boxed{\pi/12}$$

iii) $\int_1^4 \left(\frac{1}{\sqrt{t}} - 3\sqrt{t} \right) dt$

Sol

$$= \int_1^4 \frac{1}{\sqrt{t}} dt - 3 \int_1^4 \sqrt{t} dt$$

$$= 2\sqrt{t} - 2t^{3/2} \Big|_{+1}^4$$

$$= (2\sqrt{4} - 2\sqrt{1}) - [(2\sqrt{4})^{3/2} - (2\sqrt{1})^{3/2}]$$

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$$= (4 - 2) - (16 - 2)$$

$$= 2 - 14$$

$$= \boxed{-12}$$